

Xudong Zhu

Ph.D. Student in Computer Science · The Ohio State University · Columbus, OH

✉ zhu.3944@osu.edu · 🌐 xudongzhu.com · 🔗 [xzAscC](#) · 🎓 [Google Scholar](#) · 🆔 0009-0000-3068-0754

I am a Ph.D. student in Computer Science at The Ohio State University, advised by Prof. Zhihui Zhu. My research focuses on representation learning and training dynamics in large language models, using these perspectives to advance mechanistic interpretability. In particular, I study linear phenomena in model representations: how a concept propagates during generation and whether it stays linearly represented along the way, and how training methods such as SFT and RLVR shape these structures. Before OSU, I received my B.S. in Computer Science from the University of Electronic Science and Technology of China (UESTC), where I worked with Prof. Zhao Kang on document-level relation extraction and Prof. Dong Hao on graph learning.

RESEARCH INTERESTS

Representation Learning · Interpretability · Representation & Behavior Steering · Training Dynamics · Agent Behavior Control

EDUCATION

The Ohio State University

Ph.D. in Computer Science and Engineering, advised by Prof. Zhihui Zhu

2024 – Present

Columbus, OH

University of Electronic Science and Technology of China

B.S. in Computer Science · GPA 3.98 / 4.00

2020 – 2024

Chengdu, China

PUBLICATIONS

AbsTopK: Rethinking Sparse Autoencoders for Bidirectional Features

Xudong Zhu, Mohammad Mahdi Khalili, Zhihui Zhu

[ICLR 2026](#) · [Paper](#) · [Code](#) · [Project Page](#)

From Emergence to Control: Probing and Modulating Self-Reflection in Language Models

Xudong Zhu, Jiachen Jiang, Mohammad Mahdi Khalili, Zhihui Zhu

[TMLR 2026](#) · [Paper](#) · [Code](#) · [Project Page](#)

FCDS: Fusing Constituency and Dependency Syntax into Document-Level Relation Extraction

Xudong Zhu, Zhao Kang, Bei Hui

[LREC-COLING 2024](#) · [Paper](#) · [Code](#) · [Project Page](#)

Alleviating Subgraph-Induced Oversmoothing in Link Prediction via Coarse Graining

Xudong Zhu, Dong Hao, Ziqin Gao, Liming Pan

[Neurocomputing, 2025](#) · [Paper](#) · [Code](#) · [Project Page](#)

HONORS & RESEARCH SUPPORT

MMLS 2026 Travel Award, Midwest Machine Learning Symposium

2026

NAIRR Pilot Project NAIRR260106 (contributor), National AI Research Resource Pilot

Geometry and Training Dynamics of Representations in Large Language Models

2026

OPEN SOURCE

LLMUsage · Local-first LLM subscription usage monitor for Hyprland (OpenAI, GLM, Grok, Claude)

TypeScript

dotfiles · fish, Neovim, and OpenCode configuration managed with GNU Stow

Lua